Labs Guidelines

Machine Learning and Artificial Intelligence in Finance

FINC 5322

Fall 2025

The American University in Cairo

Onsi Sawiris School of Business

Heikal Department of Management

Customer Analytics & Prediction – Assignment 4

In this assignment, you will build a full customer analytics pipeline using real online retail data. You will transform raw transactions into customer-level features and build models to answer key questions:

- Who will churn?
- How long will customers stay?
- How much will they spend (CLV)?
- What customer segments exist?

You'll practice EDA, feature engineering, classification, regression, neural networks, clustering, and business interpretation.

Accessing the Assignment

- All assignments are hosted on our course Code Repository (GitHub): https://github.com/kelkess43/AUC-Material
- The assignment will also be shared weekly through Canvas modules for easy access.
- The dataset to be used is provided along with the assignment files.

Structure of the Assignment

Your notebook is divided into seven sections.
You must complete the tasks and reflections directly within the. ipynb file.
All questions, instructions, and grades are provided inside the notebook.

Important Note About the Notebook

The notebook includes guided analysis, explanations, and business context to help you understand each step of the customer analytics process.
Several cells contain commented code (# ...) that:

- Must not be removed,
- Should be used and completed, and
- Are essential for completing the assignment correctly.

These guided sections provide both technical direction and real-world business interpretation.

Section 1 – Customer-Level Feature Engineering & EDA (30 pts)

- **Q1.1 (2 pts):** Calculate the Total Price for Each Transaction
- **Q1.2 (3 pts):** Identify the Top 10 Countries by Revenue
- **Q1.3 (6 pts):** Create Customer-Level Frequency & Monetary Metrics
- **Q1.4 (4 pts):** Create the Average Invoice Value Feature
- **Q1.5 (7 pts):** Create a Churn Label
- **Q1.6 (8 pts):** Compute Customer Lifetime Value (CLV)

Section 2 – Predictive Modelling (45 pts)

- **Q2.1 (20 pts):** Build Churn Prediction Models (Logistic Regression + XGBoost)
- **Q2.2 (10 pts):** Handle Class Imbalance (Balanced Models)
- **Q2.3 (7 pts):** Tenure Prediction Using Linear Regression
- **Q2.4 (8 pts):** CLV Prediction Using a Neural Network

Section 3 – Customer Segmentation (25 pts)

- **Q3.1 (15 pts):** Full K-Means Clustering Workflow (Scaling + Elbow + Clustering)
- **Q3.2 (10 pts):** Build Final K=4 Model and Interpret Customer Segments

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Deliverables and Submission

- Submit one Jupyter Notebook (.ipynb) file with all your work.
- Make sure all cells are executed, and outputs are visible.
- Add short comments/markdown where required
- Deadline: Monday, 14th December at 11:59 PM
- Submission: Upload your completed. ipynb notebook on Canvas